AEGIS

One-page team science statement

Our starting question: how can a poultry distributor use public food-safety and environmental evidence to make better delivery decisions while keeping food available? AEGIS developed from a logistics idea into an auditable path from historical evidence to a reviewed operational choice, with a simulated Missouri network, freshness replay and driver instructions.

Complementary roles, reported by the project lead: Isaac Jenkins led the disease/public-health perspective and disease-related databases. Micky fact-checked code and helped establish the GitHub repository. The project lead handled supply-chain logistics, gathered the remaining data, defined application requirements and designed operational extensions. Codex assisted implementation, documentation and automated testing. No professional credentials or external interviews are claimed.

A decision changed: Isaac's disease-side input helped broaden the logistics concept to include CDC data, especially BEAM, and historical patterns. Human surveillance became a distinct evidence layer rather than an assumed contamination label for a shipment. The lead's feedback on confusing controls and driver distraction shaped guided questions, clearer map workflows and parked-use reporting. These are reported project decisions, not a formal usability study.

What integration taught us: FSIS supplies selected-sample outcomes, PulseNet-derived BEAM aggregates supply human context, and NOAA supplies ambient observations. NORS and USDA Food Environment Atlas add two domains. On 171 held-out Missouri samples, core Brier was 0.11246 versus baseline 0.11431; BEAM/NOAA marginal intervals included zero. NORS worsened Brier to 0.12449, with an old, constant holdout feature. More data did not automatically improve prediction. USDA remains contextual without a verified recipient/needs crosswalk.

How the idea matured: uncertain surveillance never automatically holds a lot. Binding recall/hold exclusions and human approval govern dispatch. Freshness replay and constrained comparisons explore timing, eligible volume and service; all logistics and cargo telemetry are simulated. No measured spoilage, prevented illness or real-world savings are claimed.

Next partner: a refrigerated distributor with a public-health/food-safety partner could provide deidentified traceability, actual publication timestamps, calibrated cargo histories and receiving/demand data for prospective validation and food-access analysis. Authors will review credit and claims; Micky's surname and the lead's author name remain for completion. Kiro was not used; prepared specs are not genuine Kiro evidence. Full statement: docs/team_science_statement.md.